

SSN COLLEGE OF ENGINEERING (Autonomous)
(Affiliated to Anna University, Chennai)
DEPARTMENT OF CSE

UCS 2511 NETWORK LABORATORY

A11: Simulation of Domain Name Server using UDP

Faculty In-charge: Dr.N.Sujaudeen, Assistant Professor / CSE

Batch: 2020-2024 Year: III Sem: V Sec: A

Learning Outcome:

To be proficient in developing an application to simulate domain name server using socket programming in C

Write the code using socket programming in C

Server should perform the following:

1. Maintain a DNS in the form of table. The table contains IP address and the corresponding Server name and displays the table.
2. When a request is for an IP address (Given a server name), from a client is received, verify the table and send the corresponding IP address to the client.
3. Make server to accept multiple client request simultaneously using UDP.
4. Also modify the server.

Client should do the following:

1. Request for an IP address is given to the server by the domain name.
2. Receive the corresponding IP address and display it.

Sample Input and Output

Server

Server Name	IP Address
www.yahoo.com	10.2.45.67
www.annauniv.edu	197.34.53.122
www.google.com	142.89.78.66

Do you want to modify (yes or no): yes

Domain name: www.yahoo.com

IP address: 300.8.35.79

Invalid IP address

Enter valid IP address: 197.34.53.122

IP address already exists

Enter valid IP address: 196.34.53.122

Updated table is

Server Name	IP Address
www.yahoo.com	10.2.45.67
	196.34.53.122
www.annauniv.edu	197.34.53.122
www.google.com	142.89.78.66

Client1

Enter the server name: www.annauniv.edu

The IP address is :197.34.53.122

Client2

Enter the server name: www.yahoo.edu

The IP address is :10.2.45.67

Enter the server name: www.yahoo.edu

The IP address is :10.2.45.67

196.34.53.122